



System architecture

Helios cluster is composed of two kind of partitions: CPUs and GPU partitions. Both provide a total power of 218 TFlop/s.

CPUs partitions

MPI partition

The MPI partition is composed of **16 nodes** (480 cores), based on Intel processors:

Model	PowerEdge C6420 (Dell Inc.)
Processor	Intel(R) Xeon(R) Gold 6126 CPU (Cascade Lake)
Total processor per node	2
Total cores per processor	12
Total cores per node	24
Clock rate	2.60 GHz
RAM	96 GB
Local storage (/tmp)	1 TB

SMP partition

The SMP partition is composed of **4 nodes** (128 cores), based on Intel processors:

Model	PowerEdge R740xd (Dell Inc.)
Processor	Intel(R) Xeon(R) Gold 6226R CPU (Cascade Lake)
Total processor per node	2
Total cores per processor	16
Total cores per node	32
Clock rate	2.90 GHz
RAM	96 GB
Local storage (/tmp)	1 TB

Bigmem partition

This partition is composed of **2 nodes** (112 cores) with lots of available RAM.

Firstly, there is `m3sovisu` which have 256 GB of RAM:

Model	ProLiant DL385 Gen10+ v2 (HPE)
Processor	AMD EPYC 7513 CPU (Zen3)
Total processor per node	2
Total cores per processor	32
Total cores per node	64
Clock rate	2.60 GHz

RAM	252 GB

Secondly, there is `node4-27` which have 1 TB of RAM:

Model	ProLiant DL360 Gen10 (HPE)
Processor	Intel(R) Xeon(R) Gold 6248R CPU (Cascade Lake)
Total processor per node	2
Total cores per processor	24
Total cores per node	48
Clock rate	3.00 GHz
RAM	1 TB
Local storage (/Scratch)	5.8 TB

GPU partitions

GPU partition (A100)

This partition is composed of **5 nodes** (320 cores) based on AMD processors and NVIDIA A100:

Model	PowerEdge R7525 (Dell Inc.)
Processor	AMD 7502 CPU (Zen)

Total processor per node	2
Total cores per processor	32
Total cores per node	64
Clock rate	2.50 GHz
RAM	512 GB
Local storage (/Scratch)	5.2 TB
GPUs	3x NVIDIA Tesla A100
GPU RAM	3x 40 GB

ⓘ NVIDIA A100

Mémoire : 40 Go HBM2 ECC 5120 bits (Bande passante : 1555 Go/s)

Cœurs CUDA : 6912

FP64 : 9.7 TFlops

FP32 : 19.5 TFlops

TF32 : 312 Tflops

Tensor Float 32 (TF32) : 156 TFlops

BFLOAT16 Tensor Core : 312 TFlops

FP16 Tensor Core: 1248 TFlops

INT8 Tensor Core : 624 TOPs

Jusqu'à 7 instances MIG à 5 Go

Refroidissement passif

GPU L40 partition

This partition is composed of **1 node** (124 cores) based on AMD processors and NVIDIA L40:

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Model	ProLiant DL385 Gen11 (HPE)
Processor	AMD EPYC 9534 CPU (Zen4)
Total processor per node	2
Total cores per processor	64
Total cores per node	128
Clock rate	2.45 GHz
RAM	1 TB
Local storage (/Scratch)	3.5 TB
GPUs	2x NVIDIA L40
GPU RAM	2x 48 GB

! NVIDIA L40

Mémoire : 48GB GDDR6 with ECC (Bande passante: 864 GB/s)

Cœurs CUDA : 18 176

NVIDIA third-generation RT cores : 142

NVIDIA fourth-generation Tensor cores : 568

RT Core performance : 209 TFlops FP32 : 90.5 TFlops

TF32 : 181 TFlops

BFLOAT16 Tensor Core : 362.1 TFlops

FP16 Tensor Core : 362.1 TFlops

FP8 Tensor Core : 724 TFlops

INT8 Tensor Core : 724 TOPs

Refroidissement passif

Login node

Model	PowerEdge R940 (Dell Inc.)
Processor	Intel(R) Xeon(R) Platinum 8160 CPU (Cascade Lake)
Total cores per processor	24
Total cores per node	96
Clock rate	2.60 GHz
RAM	256 GB
Local storage (/tmp)	1 TB

Networks

- 100G/s Omnipath for MPI and RDMA applications. **Available only on CPU partition.**
- 10 G/s Ethernet for Data transfer and network file systems (NFS, BeeGFS)

Operating System

- Rocky Linux 8.5 is deployed on all nodes
- Possibility to use Singularity/Apptainer containers

Storage

- 250 TB (BeeGFS and NFS) over 10 G/s network

Slurm partition organization

! INFO

Detailed information of Slurm partitions can be found [here](#).